

L2 - 9.3. Atomic Red Team

Task 1 - Introduction

How do threat actors execute initial payloads? What typical commands are performed by threat actors once a persistent connection is established in the network? What does it look like in our environment?

Task 2 - Atomic Red Team

It is an open-source project that provides a framework for performing security testing and threat emulation. It consists of tools and techniques that can be used to simulate various types of attacks and security threats, such as malware, phishing attacks, and network compromise. The Atomic Red Team aims to help security professionals assess the effectiveness of their organization's security controls and incident response processes and identify areas for improvement.

The Atomic Red Team framework is designed to be modular and flexible, allowing security professionals to select the tactics and techniques most relevant to their testing needs. It is intended to be used with other tools and frameworks, such as the MITRE ATT&CK framework, which provides a comprehensive overview of common tactics and techniques threat actors use.

Supported Platforms

Platform Type	Platforms Supported
Operating System	Windows, Linux, macOS
Cloud Infrastructure	AWS, Azure, GCP
Cloud Services	Office 365, Google Workspaces, Azure AD
Others	Containers (Kubernetes)

Understanding How Emulation Works: Atomic Red Team emulates commands that mimic threat activity using **Executors**. Below is the list of available executors.

Executor	Operating System	Notes
sh or bash - /bin/sh or /bin/bash	Linux, macOS	Commands executed by this Executor are usually Unix tools used by threat actors for malicious intent.

Command Prompt - <code>cmd.exe</code>	Windows	Commands executed by this Executor are usually Windows Built-in or Third-party binaries used by threat actors for malicious intent.
PowerShell - <code>powershell.exe</code>	Windows	Emulated commands by this Executor are commonly known malicious PowerShell modules that threat actors abuse.
Manual	N/A	The details given in this type are typically written as steps needed to be executed to emulate a threat, such as when GUI steps are involved that cannot be automated.

Atomics refers to different testing techniques based on the MITRE ATT&CK Framework. Each works as a standalone testing mock-up that Security Analysts can use to emulate a specific Technique, such as **OS Credential Dumping: LSASS Memory**, for a quick example. Each Atomic typically contain two files, both of which are named by their MITRE ATT&CK Technique ID:

T1003.001 - OS Credential Dumping: LSASS Memory

```
user@ATOMIC$ ls -lh T1003.001/
-rw-r--r--  1 user  user      300B Jan  4 22:57 T1003.001.md
-rw-r--r--  1 user  user      500B Jan  4 22:58 T1003.001.yaml
```

- **Markdown File (.md)** - Contains all the information about the technique, the supported platform, Executor, GUID, and commands to be executed.
- **YAML File (.yaml)** - Configuration used by frameworks, such as Invoke-Atomic and Atomic-Operator, to do the exact emulation of the technique

The Markdown file is written to be self-explanatory, so let's dive deep into the configuration files used to emulate commands. *There are 12 atomic tests for T1003.001, but we will present only one test below for brevity.*

Atomic YAML File Breakdown

The first few fields are already given based on their field names:

- **attack_technique** - MITRE ATT&CK Technique ID, which also signifies the file's name.
- **display_name** - The technique name, similar to how it is presented as a MITRE Technique.
- **atomic_tests** - List of atomic tests, which details how every test is executed.

T1003.001 - OSCredential Dumping: LSASS Memory

```
user@ATOMIC$ cat T1003.001/T1003.001.yaml

attack_technique: T1003.001
display_name: "OS Credential Dumping: LSASS Memory"
atomic_tests:
  ...
```

The following section details the contents of a single Atomic Test under the list of **atomic_tests** field:

- **name** Short snippet that describes how it tests the technique.
- **auto_generated_guid** Unique identifier of the specific test.
- **description** A longer form of the test details and can be written in a multi-line format.
- **supported_platforms** On what platform will the technique be executed (on a Windows machine in this case)
- **input_arguments** Required values during the execution, resorts to the default value if nothing is supplied.

T1003.001 - OSCredential Dumping: LSASS Memory

```
...
- name: Dump LSASS.exe Memory using ProcDump
  auto_generated_guid: 0be2230c-9ab3-4ac2-8826-3199b9a0ebf8
  description: |
    The memory of lsass.exe is often dumped for offline credential
    theft attacks. This can be achieved with Sysinternals
    ProcDump.
```

Upon successful execution, you should see the following file created c:\windows\temp\lsass_dump.dmp.

If you see a message saying "procdump.exe is not recognized as an internal or external command", try using the get-prereq_commands to download and install the ProcDump tool first.

```
supported_platforms:
```

```
- windows
input_arguments:
  output_file:
    description: Path where resulting dump should be placed
    type: Path
    default: C:\Windows\Temp\lsass_dump.dmp
  procdump_exe:
    description: Path of Procdump executable
    type: Path
    default: PathToAtomicsFolder\T1003.001\bin\procdump.exe
```

To conclude with the contents of an Atomic test, details about **dependencies** and **executors** are as follows:

- **dependency_executor_name** - Option on how the prerequisites will be validated. The possible values for this field are similar to the **Executor** field.
- **dependencies**
 - **prereq_command** Commands to check if the requirements for running this test are met. The conditions for the "command_prompt" Executor are not satisfied if any command returns a non-zero exit code. For the "Powershell" Executor, all commands are run as a script block, and the script block must return 0 for success.
 - **get_prereq_command** Commands to meet this prerequisite or a message describing how to meet this requirement.
- **executor**
 - **name** - Name of the Executor; similar to what has been discussed above.
 - **command** - Exact command to emulate the technique.
 - **cleanup_command** Commands for cleaning up the previous atomic test, such as deletion of files or reverting modified configurations.
 - **elevation_required** A boolean value that dictates if an admin privilege is required.

T1003.001 - OS Credential Dumping: LSASS Memory

```
...
dependency_executor_name: powershell
```

```

dependencies:
- description: |
    ProcDump tool from Sysinternals must exist on disk at specified location ({procdump_exe})
  prereq_command: |
    if (Test-Path {procdump_exe}) {exit 0} else {exit 1}
  get_prereq_command: |
    [Net.ServicePointManager]::SecurityProtocol = [Net.SecurityProtocolType]::Tls12
    Invoke-WebRequest "https://download.sysinternals.com/files/Procdump.zip" -OutFile "$env:TEMP\Procdump.zip"
    Expand-Archive $env:TEMP\Procdump.zip $env:TEMP\Procdump -Force
    New-Item -ItemType Directory (Split-Path {procdump_exe}) -Force | Out-Null
    Copy-Item $env:TEMP\Procdump\Procdump.exe {procdump_exe} -Force
  executor:
    name: command_prompt
    command: |
      {procdump_exe} -accepteula -ma lsass.exe {output_file}
    cleanup_command: |
      del "{output_file}" >nul 2> nul
    elevation_required: true

```

Before concluding this section, it is essential to highlight the **cleanup** feature of Atomics. But why is cleanup significant in threat emulation?

- First and foremost, it helps to ensure that the tested system returns to its original state after completing the emulation.
- Next, writing cleanup scripts help in minimizing the risk of accidental damage to the system since it serves as an additional verification of the execution. You understand the impact more as you write the reverse of the emulation.
- Lastly, it helps to maintain the integrity of the system being tested. Tools left after emulation exercises could be abused by potential threat actors.

integrity

Answer the questions below

- What type of executor is used for actions that cannot be automated? manual
- What is the field in an Atomic YAML file populated by a unique identifier to isolate a specific Atomic? auto_generated_guid
- What is the field in an Atomic YAML file populated by commands for deleting files used for emulation or reverting modified configurations? cleanup_command

Task 3 - Invoke-AtomicRedTeam

The main goal of the Atomic Red Team Framework is to ease the threat emulation process. Hence, tools such as **Invoke-AtomicTest** were created to utilise the automation and execution of Atomics. Let's start playing with these tools by emulating different attack patterns.

[Invoke-AtomicRedTeam](#) (opens in new tab) is a PowerShell module created by the same author (Red Canary) that allows Security Analysts to run simulations defined by Atomics. To avoid confusion, the primary cmdlet used in this module is `Invoke-AtomicTest` and not `Invoke-AtomicRedTeam`.

Listing Atomic Techniques:

Administrator: Windows PowerShell

```
PS C:\Users\Administrator> Invoke-AtomicTest T1127 -ShowDetailsBrief
PathToAtomicsFolder = C:\Tools\AtomicRedTeam\atomics

T1127-1 Lolbin Jsc.exe compile javascript to exe
T1127-2 Lolbin Jsc.exe compile javascript to dll
```

The output shows that **ShowDetailsBrief** lists the available tests in the specified Atomic and its corresponding Atomic Test number.

Administrator: Windows PowerShell

```
PS C:\Users\Administrator> Invoke-AtomicTest T1127 -ShowDetails
PathToAtomicsFolder = C:\Tools\AtomicRedTeam\atomics

[*****BEGIN TEST*****]
Technique: Trusted Developer Utilities Proxy Execution T1127
Atomic Test Name: Lolbin Jsc.exe compile javascript to exe
```

Atomic Test Number: 1
Atomic Test GUID: 1ec1c269-d6bd-49e7-b71b-a461f7fa7bc8
Description: Use jsc.exe to compile javascript code stored in scriptfile.js and output scriptfile.exe. <https://lolbas-project.github.io/lolbas/Binaries/Jsc/> <https://www.phpied.com/make-your-javascript-a-windows-exe/>

Attack Commands:

Executor: command_prompt

ElevationRequired: False

Command:

```
copy #{filename} %TEMP%\hello.js  
#{jscpath}\#{jscname} %TEMP%\hello.js
```

Command (with inputs):

```
copy C:\Tools\AtomicRedTeam\atomics\T1127\src\hello.js %TEMP%\hello.js  
C:\Windows\Microsoft.NET\Framework\v4.0.30319\jsc.exe %TEMP%\hello.js
```

Cleanup Commands:

Command:

```
del %TEMP%\hello.js  
del %TEMP%\hello.exe
```

---- Dependencies section is redacted, and will be discussed in the next section ---

The output above shows that the ShowDetails parameter is the verbose version of ShowDetailsBrief, which only provides the list of tests inside the Atomic T1127.

Preparing Atomic Prerequisites

Administrator: Windows PowerShell

Dependencies:

Description: JavaScript code file must exist on disk at specified location (C:\Tools\AtomicRedTeam\atomics\T1127\src\hello.js)

Check Prereq Command:

```
if (Test-Path #{filename}) {exit 0} else {exit 1}
```

Check Prereq Command (with inputs):

```

if (Test-Path C:\Tools\AtomicRedTeam\atomics\T1127\src\hello.js)
{exit 0} else {exit 1}
Get Prereq Command:
New-Item -Type Directory (split-path #{filename}) -ErrorAction ignore | Out-Null
Invoke-WebRequest "https://github.com/redcanaryco/atomic-red-team/raw/master/atomics/T1127/src/hello.js" -OutFile "#{filename}"
Get Prereq Command (with inputs):
New-Item -Type Directory (split-path C:\Tools\AtomicRedTeam\atomics\T1127\src\hello.js) -ErrorAction ignore | Out-Null
Invoke-WebRequest "https://github.com/redcanaryco/atomic-red-team/raw/master/atomics/T1127/src/hello.js" -OutFile "C:\Tools\AtomicRedTeam\atomics\T1127\src\hello.js"
[!!!!!!!END TEST!!!!!!!]

```

It is crucial to verify if the dependencies are met before executing the tests, and this can be done by using the **CheckPrereqs** parameter. In the case of **Atomic T1127-1**, the file from **C:\Tools\AtomicRedTeam\atomics\T1127\src\hello.js** should exist.

Administrator: WindowsPowerShell

```

PS C:\Users\Administrator> Invoke-AtomicTest T1127 -CheckPrereqs
PathToAtomicsFolder = C:\Tools\AtomicRedTeam\atomics

```

```

CheckPrereq's for: T1127-1 Lolbin Jsc.exe compile javascript to exe
Prerequisites met: T1127-1 Lolbin Jsc.exe compile javascript to exe
CheckPrereq's for: T1127-2 Lolbin Jsc.exe compile javascript to dll
Prerequisites met: T1127-2 Lolbin Jsc.exe compile javascript to dll

```

```

PS C:\Users\Administrator> Invoke-AtomicTest T1127 -GetPrereqs
PathToAtomicsFolder = C:\Tools\AtomicRedTeam\atomics

```

```

GetPrereq's for: T1127-1 Lolbin Jsc.exe compile javascript to exe

```

Attempting to satisfy prereq: JavaScript code file must exist on disk at specified location (C:\Tools\AtomicRedTeam\atomics\T1127\src\hello.js)

Prereq already met: JavaScript code file must exist on disk at specified location (C:\Tools\AtomicRedTeam\atomics\T1127\src\hello.js)

GetPrereq's for: T1127-2 Lolbin Jsc.exe compile javascript to dll

Attempting to satisfy prereq: JavaScript code file must exist on disk at specified location (C:\Tools\AtomicRedTeam\atomics\T1127\src\LibHello.js)

Prereq already met: JavaScript code file must exist on disk at specified location (C:\Tools\AtomicRedTeam\atomics\T1127\src\LibHello.js)

There are multiple ways to execute the Atomic tests, which will be detailed below.

Parameter	Example	Details
TestNumbers	<code>Invoke-AtomicTest T1127 -TestNumbers 1,2</code>	Executes tests based on the Atomic test number
TestNames	<code>Invoke-AtomicTest T1127 -TestNames "Lolbin Jsc.exe compile javascript to dll"</code>	Executes tests based on the Atomic test names
TestGuids	<code>Invoke-AtomicTest T1127 -TestGuids 3fc9fea2-871d-414d-8ef6-02e85e322b80</code>	Executes tests based on the test GUID
N/A	<code>Invoke-AtomicTest T1127</code>	Executes all tests under Atomic T1127
N/A	<code>Invoke-AtomicTest T1127-2</code>	Executes Atomic Test #2 of T1127

Answer the questions below

Checked if atomic test works:

```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\Administrator> powershell -ExecutionPolicy bypass
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\Administrator> Import-Module "C:\Tools\invoke-atomicredteam\Invoke-AtomicRedTeam.psdl" -Force
PS C:\Users\Administrator> $PSDefaultParameterValues = @{ "Invoke-AtomicTest:PathToAtomicsFolder"="C:\Tools\AtomicRedTeam\atomics" }
PS C:\Users\Administrator> help Invoke-AtomicTest

NAME
    Invoke-AtomicTest

SYNTAX
    Invoke-AtomicTest [-AtomicTechnique] <string[]> [-ShowDetails] [-ShowDetailsBrief] [-TestNumbers <string[]>]
    [-TestNames <string[]>] [-TestGuids <string[]>] [-PathToAtomicsFolder <string>] [-CheckPrereqs]
    [-PromptForInputArgs] [-GetPrereqs] [-Cleanup] [-NoExecutionLog] [-ExecutionLogPath <string>] [-Force] [-InputArgs
    <hashtable>] [-TimeoutSeconds <int>] [-Session <PSSession[]>] [-Interactive] [-KeepStdOutStdErrFiles]
    [-LoggingModule <string>] [-WhatIf] [-Confirm] [<CommonParameters>]

ALIASES
    None

REMARKS
    None

PS C:\Users\Administrator>
```

How many atomic tests are under Atomic T1110.001 that are supported on Windows hosts? 4

Invoke-AtomicTest T1110.001 -ShowDetailsBrief

What is the Atomic name of the second test under Atomic T1218.005?

Invoke-AtomicTest T1218.005 -ShowDetailsBrief

Mshta executes VBScript to execute malicious command

How many prerequisites are not met for Atomic T1003? 4

Invoke-AtomicTest T1003 -CheckPrereqs

4 prereqs are not met

What is the parameter used to execute the specific Atomic test via GUID?

TestGuids	Invoke-AtomicTest T1127 -TestGuids 3fc9fea2-871d-414d-8ef6-02e85e322b80
-----------	---

What is the name of the scheduled task created after executing the 2nd test of Atomic T1053.005? spawn

Invoke-AtomicTest T1053.005 -ShowDetailsBrief

(then) Invoke-AtomicTest T1053.005 -TestNumbers 2

```
PS C:\Users\Administrator> Invoke-AtomicTest T1053.005 -ShowDetailsBrief
PathToAtomicsFolder = C:\Tools\AtomicRedTeam\atomics

T1053.005-1 Scheduled Task Startup Script
T1053.005-2 Scheduled task Local
T1053.005-3 Scheduled task Remote
T1053.005-4 Powershell Cmdlet Scheduled Task
T1053.005-5 Task Scheduler via VBA
T1053.005-6 WMI Invoke-CimMethod Scheduled Task
T1053.005-7 Scheduled Task Executing Base64 Encoded Commands From Registry
T1053.005-8 Import XML Schedule Task with Hidden Attribute
T1053.005-9 PowerShell Modify A Scheduled Task
PS C:\Users\Administrator> Invoke-AtomicTest T1053.005 -TestNumbers 2
PathToAtomicsFolder = C:\Tools\AtomicRedTeam\atomics

Executing test: T1053.005-2 Scheduled task Local
SUCCESS: The scheduled task "spawn" has successfully been created.
Done executing test: T1053.005-2 Scheduled task Local
PS C:\Users\Administrator>
```

What is the registry key modified after executing the 2nd test of Atomic T1547.001?

Invoke-AtomicTest T1547.001 -ShowDetailsBrief

```

PS C:\Users\Administrator> Invoke-AtomicTest T1547.001 -ShowDetailsBrief
PathToAtomicsFolder = C:\Tools\AtomicRedTeam\atomics

T1547.001-1 Reg Key Run
T1547.001-2 Reg Key RunOnce
T1547.001-3 PowerShell Registry RunOnce
T1547.001-4 Suspicious vbs file run from startup Folder
T1547.001-5 Suspicious jse file run from startup Folder
T1547.001-6 Suspicious bat file run from startup Folder
T1547.001-7 Add Executable Shortcut Link to User Startup Folder
T1547.001-8 Add persistence via Recycle bin
T1547.001-9 SystemBC Malware-as-a-Service Registry
T1547.001-10 Change Startup Folder - HKLM Modify User Shell Folders Common Startup Value
T1547.001-11 Change Startup Folder - HKCU Modify User Shell Folders Startup Value
T1547.001-12 HKCU - Policy Settings Explorer Run Key
T1547.001-13 HKLM - Policy Settings Explorer Run Key
T1547.001-14 HKLM - Append Command to Winlogon Userinit KEY Value
T1547.001-15 HKLM - Modify default System Shell - Winlogon Shell KEY Value
T1547.001-16 secedit used to create a Run key in the HKLM Hive
PS C:\Users\Administrator> Invoke-AtomicTest T1547.001 -TestNumbers 2
PathToAtomicsFolder = C:\Tools\AtomicRedTeam\atomics

Executing test: T1547.001-2 Reg Key RunOnce
The operation completed successfully.
Done executing test: T1547.001-2 Reg Key RunOnce
PS C:\Users\Administrator>

```

Invoke-AtomicTest T1547.001 -ShowDetails

HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\RunOnceEx\0001\Depend

```

[*****BEGIN TEST*****]
Technique: Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder T1547.001
Atomic Test Name: Reg Key RunOnce
Atomic Test Number: 2
Atomic Test GUID: 554cbd88-cde1-4b56-8168-0be552eed9eb
Description: RunOnce Key Persistence.
Upon successful execution, cmd.exe will modify the registry to load AtomicRedTeam.dll to
tdout.

Attack Commands:
Executor: command_prompt
ElevationRequired: True
Command:
REG ADD HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\RunOnceEx\0001\Depend /v 1 /d "C:\
Command (with inputs):
REG ADD HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\RunOnceEx\0001\Depend /v 1 /d "C:\

```

then cleanup

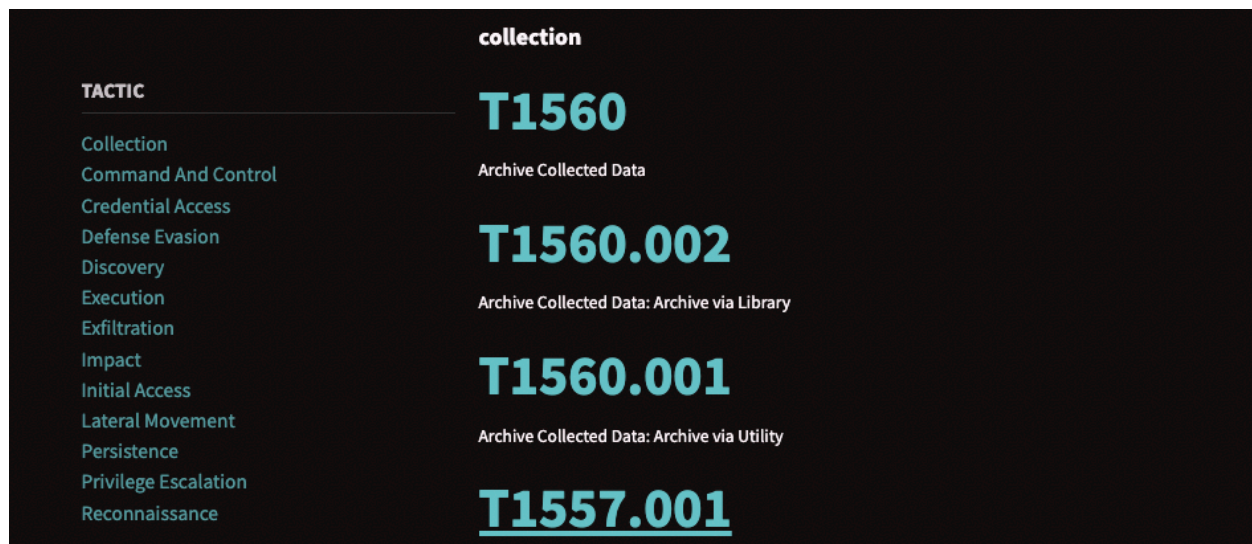
Invoke-AtomicTest T1547.001 -TestNumbers 2 -Cleanup

Task 4 - Revisiting MITRE ATT&CK

Import-Module "C:\Tools\invoke-atomicredteam\Invoke-AtomicRedTeam.psd1" - Force

\$PSDefaultParameterValues = @{"Invoke-AtomicTest:PathToAtomicsFolder"="C:\Tools\AtomicRedTeam\atomics"}

Based on the previous tasks, it was highlighted that the Atomic Red Team framework is heavily tied-up with MITRE ATT&CK. One clear connection is that every Atomic is written for a specific MITRE Technique, with the files of every Atomic named as its corresponding MITRE Technique ID. In addition, an excerpt from atomicredteam.io ([opens in new tab](#)) shows the available Tactics and Atomics under it, the Collection tactic in this case.



This correlation shows us that to maximise the Atomic Red Team framework's usage, we need to utilise our knowledge in MITRE ATT&CK Framework. Let's play again with the ATT&CK Navigator, one of the tools introduced in the [Threat Modeling room](#).

ATT&CK Navigator

Incorporating ATT&CK Navigator into Atomic Red Team

Choosing what Atomic to pick might be overwhelming; thus, we need to have a sense of direction when doing threat emulation. By utilising ATT&CK Navigator, we can dump the TTPs used by known APTs or cybercriminal groups and emulate their Atomics.

To start, create a new layer in the ATT&CK Navigator and choose **Enterprise**.

MITRE ATT&CK® Navigator

The ATT&CK Navigator is a web-based tool for annotating and exploring ATT&CK matrices. It can be used to visualize defensive coverage, red/blue team planning, the frequency of detected techniques, and more.

help changelog theme ▼

Create New Layer
Create a new empty layer

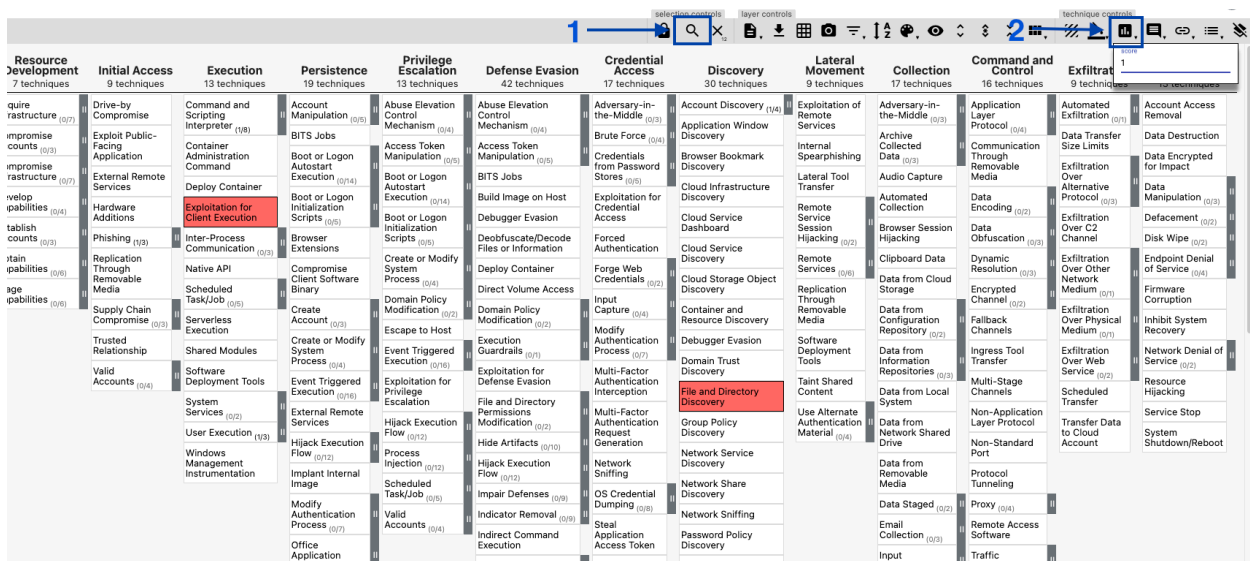
Enterprise
Mobile
ICS

More Options

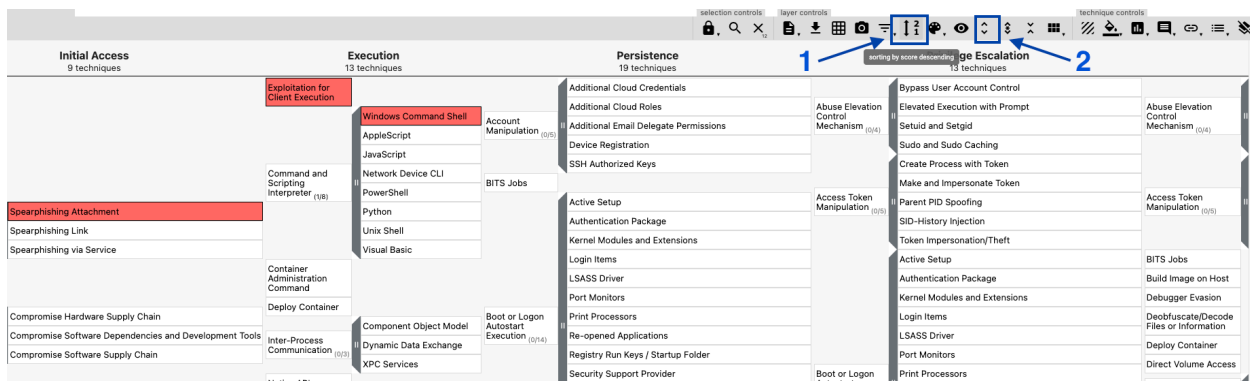
Then use the search functionality and navigate to the threat groups section. Let's select **admin@338** as our interest group by clicking the select button. This action sets the techniques attributed to the **admin@338** group.

The screenshot shows the MITRE ATT&CK Navigator interface. On the right, a search sidebar is open, displaying 'Techniques (594)' and 'Threat Groups (133)'. In the 'Threat Groups' section, the group 'admin@338' is selected, indicated by a blue highlight and a 'select' button. A blue arrow labeled '2' points to this selection. The main area displays a grid of ATT&CK techniques, categorized into columns such as 'Resource Development', 'Initial Access', 'Execution', 'Persistence', 'Privilege Escalation', 'Defense Evasion', 'Credential Access', 'Discovery', and 'Lateral Movement'. A blue arrow labeled '1' points to the search icon in the top right corner of the main grid.

Next, close the search sidebar by re-clicking the search icon, click the score button in the upper-right corner and put one (1) as its score. This action highlights all selected techniques, which eases out the readability of the matrix.



Lastly, let's sort out the techniques per column to place the scored technique in the topmost position. Click the filter button until it is set to **sorting by score descending**, and eventually click the **expand sub-techniques** button to display all highlighted techniques and sub-techniques.



Our steps should reflect all techniques we want to emulate from our selected threat group.

Emulation of Admin@338 Group

The following action is to list all techniques highlighted from the ATT&CK Navigator. It is essential to take note of the Tactic, Technique and its corresponding Technique ID, as every Atomic is named with the Technique ID. You may hover the mouse icon in the techniques to see the technique ID. Below is the summarised results we got from ATT&CK Navigator.

Tactic	Technique ID	Technique
--------	--------------	-----------

Initial Access	T1566.001	Spearphishing Attachment
Execution	T1203	Exploitation for Client Execution
Execution	T1059.003	Command and Scripting Interpreter: Windows Command Shell
Discovery	T1083	File and Directory Discovery
Discovery	T1082	System Information Discovery
Discovery	T1016	System Network Configuration Discovery
Discovery	T1049	System Network Connections Discovery
Discovery	T1007	System Service Discovery
Discovery	T1087.001	Account Discovery: Local Account

We have extracted nine (9) out of all the techniques in MITRE ATT&CK. The next thing we need to do is to determine if an Atomic exists for these techniques.

Administrator: WindowsPowerShell

```
PS C:\Users\Administrator>ls C:\Tools\AtomicRedTeam\atomics | Where-Object Name -Match "T1566.001|T1203|T1059.003|T1083|T1082|T1016|T1049|T1007|T1087.001"
```

Directory: C:\Tools\AtomicRedTeam\atomics

Mode	LastWriteTime	Length	Name
d-----	1/3/2023 5:20 PM		T1007
d-----	1/3/2023 5:20 PM		T1016
d-----	1/3/2023 5:20 PM		T1049
d-----	1/3/2023 5:20 PM		T1059.003
d-----	1/3/2023 5:20 PM		T1082
d-----	1/3/2023 5:20 PM		T1083
d-----	1/3/2023 5:20 PM		T1087.001
d-----	1/3/2023 5:21 PM		T1566.001

Eight out of nine techniques are available to be emulated. Before executing the tests, we must verify each technique and list all known tests for each Atomic.

Given the number of tests for each Atomic, as shown above, we will only select one for brevity. Let's also focus on the tests that have met the prerequisites.

- **T1059.003-3** - Suspicious Execution via Windows Command Shell
- **T1083-1** - File and Directory Discovery (cmd.exe)
- **T1082-6** - Hostname Discovery (Windows)
- **T1016-1** - System Network Configuration Discovery on Windows
- **T1049-1** - System Network Connections Discovery
- **T1007-2** - System Service Discovery - net.exe
- **T1087.001-9** - Enumerate all accounts via PowerShell (Local)
- **T1566.001-1** - Download Macro-Enabled Phishing Attachment

Answer the questions below

Using the ATT&CK Navigator, how many techniques are attributed to admin@338?

navigate mitre 9

The screenshot shows the ATT&CK Navigator interface. The top navigation bar includes 'layer', 'selection controls', 'layer controls', and 'technique controls'. The main content area is divided into four columns: Reconnaissance (10 techniques), Resource Development (7 techniques), Initial Access (9 techniques), and Execution (12 techniques). The right sidebar contains a search bar with 'admin@338' entered, search settings (name, ATT&CK ID, description, data sources), and results for Threat Groups (1) and Software (3). The Threat Groups section shows 'admin@338' with a 'view' link and 'select'/'deselect' buttons. The Software section shows 'BUBBLEWRAP' with a 'view' link and 'select'/'deselect' buttons.

Reconnaissance	Resource Development	Initial Access	Execution
10 techniques	7 techniques	9 techniques	12 techniques
Active Scanning (0/3)	Acquire Infrastructure (0/6)	Drive-by Compromise	Command and Scripting Interpreter (0/8)
Gather Victim Host Information (0/4)	Compromise Accounts (0/2)	Exploit Public-Facing Application	Container Administration Command
Gather Victim Identity Information (0/3)	Compromise Infrastructure (0/5)	External Remote Services	Deploy Container
Gather Victim Network Information (0/6)	Develop Capabilities (0/4)	Hardware Additions	Exploitation for Client Execution
Gather Victim Org Information (0/4)	Establish Accounts (0/2)	Phishing (0/3)	Inter-Process Communication
Phishing for Information (0/3)	Obtain Capabilities (0/6)	Replication Through Removable Media	Native API
Search Closed Sources (0/2)	Stage Capabilities (0/5)	Supply Chain Compromise (0/3)	Scheduled Task/Job (0/5)
Search Open Technical Databases (0/5)		Trusted Relationship	Shared Modules
Search Open Websites/Domains (0/2)		Valid Accounts (0/4)	Software Deployment Tool
Search Victim-Owned Websites			System Services (0/2)
			User Execution (0/4)
			Windows Management Instrumentation

Using the mapping provided by the ATT&CK Navigator, what is the Technique ID of the phishing technique used by the threat group? Given the list up T1566.001

T1007-2 - System Service Discovery - net.exe

T1087.001-9 - Enumerate all accounts via PowerShell (Local)

T1566.001-1 - Download Macro-Enabled Phishing Attachment

How many Atomic tests on Atomic T1083 are supported on Windows hosts? 4

```
PS C:\Users\Administrator> Invoke-AtomicTest T1083 -ShowDetailsBrief
PathToAtomicsFolder = C:\Tools\AtomicRedTeam\atomics
```

T1083-1 File and Directory Discovery (cmd.exe)

T1083-2 File and Directory Discovery (PowerShell)

T1083-5 Simulating MAZE Directory Enumeration

T1083-6 Launch DirLister Executable

What file should exist to satisfy the prerequisite of Atomic Test T1049-4?

Sharpview.exe

```
'T1049' | ForEach-Object {echo "Enumerating $_"; Invoke-AtomicTest $_ -CheckPrereqs }
```

```
PS C:\Users\Administrator> 'T1049-4' | ForEach-Object {echo "Enumerating $_"; Invoke-AtomicTest $_ -CheckPrereqs }
Enumerating T1049-4
PathToAtomicsFolder = C:\Tools\AtomicRedTeam\atomics

CheckPrereq's for: T1049-4 System Discovery using SharpView
Prerequisites not met: T1049-4 System Discovery using SharpView
[*] Sharpview.exe must exist on disk at specified location (C:\Tools\AtomicRedTeam\atomics\T1049\bin\SharpView.exe)

Try installing prereq's with the -GetPrereqs switch
PS C:\Users\Administrator> _
```

What is the echoed string upon executing Atomic Test T1059.003-3? Hello, from CMD!

```
Invoke-AtomicTest T1059.003-3 -ShowDetails
```

```

PS C:\Users\Administrator> Invoke-AtomicTest T1059.003-3 -ShowDetails
PathToAtomicsFolder = C:\Tools\AtomicRedTeam\atomics

[*****BEGIN TEST*****]
Technique: Command and Scripting Interpreter: Windows Command Shell T1059.003
Atomic Test Name: Suspicious Execution via Windows Command Shell
Atomic Test Number: 3
Atomic Test GUID: d0eb3597-a1b3-4d65-b33b-2cda8d397f20
Description: Command line executed via suspicious invocation. Example is from the 2021 Threat Detection Report by Red Canary.

Attack Commands:
Executor: command_prompt
ElevationRequired: False
Command:
%LOCALAPPDATA%\~3,1%md /c echo #{input_message} > #{output_file} & type #{output_file}
Command (with inputs):
%LOCALAPPDATA%\~3,1%md /c echo Hello, from CMD! > hello.txt & type hello.txt
[!!!!!!END TEST!!!!!!]

```

What is the hostname of the machine based on Atomic Test T1082-6? ATOMIC

Invoke-AtomicTest T1082-6 -ShowDetails

Invoke-AtomicTest T1082-6

Invoke-AtomicTest T1082-6 -Cleanup

```

PS C:\Users\Administrator> Invoke-AtomicTest T1082-6
PathToAtomicsFolder = C:\Tools\AtomicRedTeam\atomics

Executing test: T1082-6 Hostname Discovery (Windows)
ATOMIC
Done executing test: T1082-6 Hostname Discovery (Windows)
PS C:\Users\Administrator>

```

How many accounts are disabled based on Atomic Test T1087.001-9?

Invoke-AtomicTest T1087.001-9

```

The command completed successfully.
Name      Enabled Description
-----
Administrator      True   Built-in account for administering the computer/domain
DefaultAccount      False  A user account managed by the system.
Guest               False  Built-in account for guest access to the computer/domain
WDAGUtilityAccount False  A user account managed and used by the system for Windows Defender Application Guard
scen...

```

Task 5 - Emulation to Detection

The benefits of threat emulation in detection engineering.

Observing Telemetry: The easiest way to understand different TTPs is to see them first-hand. For this section, let's use the tests from Atomic T1547.001, which is all about Boot or Logon Autostart Execution: Registry Run Keys / Startup FolderPermalink.

Administrator: WindowsPowerShell

```
PS C:\Users\Administrator> Invoke-AtomicTest T1547.001 -CheckPrereqs
PathToAtomicsFolder = C:\Tools\AtomicRedTeam\atomics

CheckPrereq's for: T1547.001-1 Reg Key Run
Prerequisites met: T1547.001-1 Reg Key Run
CheckPrereq's for: T1547.001-2 Reg Key RunOnce
Prerequisites met: T1547.001-2 Reg Key RunOnce
CheckPrereq's for: T1547.001-3 PowerShell Registry RunOnce
Prerequisites met: T1547.001-3 PowerShell Registry RunOnce
CheckPrereq's for: T1547.001-4 Suspicious vbs file run from startup Folder
Prerequisites met: T1547.001-4 Suspicious vbs file run from startup Folder
CheckPrereq's for: T1547.001-5 Suspicious jse file run from startup Folder
Prerequisites met: T1547.001-5 Suspicious jse file run from startup Folder
CheckPrereq's for: T1547.001-6 Suspicious bat file run from startup Folder
Prerequisites met: T1547.001-6 Suspicious bat file run from startup Folder
CheckPrereq's for: T1547.001-7 Add Executable Shortcut Link to User Startup
```

There are sixteen available tests for this Atomic, and all have prerequisites met. All of the tests can be executed without any problems.

Sysmon: For this task, we will view logs generated by Atomics through Sysmon. Let's fire up the Event Viewer and navigate to `Applications and Services > Microsoft > Windows > Sysmon`. Once done, clear the existing records and start executing the first test.

Administrator: Windows PowerShell

```
PS C:\Users\Administrator> Invoke-AtomicTestT1547.001 -TestNumbers 1
PathToAtomicsFolder = C:\Tools\AtomicRedTeam\atomics
```

Executing test: T1547.001-1 Reg Key Run
The operation completed successfully
Done executing test: T1547.001-1 Reg Key Run

Refresh the Event Viewer to view the latest updates, and always clear the logs before executing the next Atomic test.

Operational Number of events: 5				
Level	Date and Time	Source	Event ID	Task Category
Information	1/9/2023 1:51:45 PM	Sysmon	13	Registry value set (rule: Re...
Information	1/9/2023 1:51:45 PM	Sysmon	1	Process Create (rule: Proce...
Information	1/9/2023 1:51:45 PM	Sysmon	1	Process Create (rule: Proce...
Information	1/9/2023 1:51:45 PM	Sysmon	1	Process Create (rule: Proce...
Information	1/9/2023 1:51:45 PM	Sysmon	1	Process Create (rule: Proce...

Note: Log clearing will only be practical if the emulation is done in a test environment. You may not have the luxury of clearing the logs in a production setup. This may require an additional understanding of the Atomic breakdown before executing it to know what logs are expected to appear.

Upon completing Atomic Test **T1547.001-1**, it generated five records. We can ignore the first two logs as it is part of **Invoke-AtomicTest**'s execution; hence we can overlook the execution of the following binaries with PowerShell being its parent process throughout this task:

- whoami.exe
- hostname.exe

ProcessId: 1160
Image: C:\Windows\System32\whoami.exe
FileVersion: 10.0.17763.1 (WinBuild.160101.0800)
Description: whoami - displays logged on user information
Product: Microsoft® Windows® Operating System
Company: Microsoft Corporation
OriginalFileName: whoami.exe
CommandLine: "C:\Windows\system32\whoami.exe"
CurrentDirectory: C:\Users\Administrator\
User: ATOMIC\Administrator
LogonGuid: {c5d2b969-17ff-63bc-42a1-070000000000}
LogonId: 0x7A142
TerminalSessionId: 2
IntegrityLevel: High
Hashes: MD5=43C2D3293AD939241DF61B3630A9D3B6,SHA256=1D5491E3C468EE4B4EFEDFF4BBC7D06EE83180F60B1576763EA2EFE049493A,IMPHASH=7FF0758B766F747CE57DFAC70743FB88
ParentProcessGuid: {c5d2b969-1967-63bc-9f00-000000001b01}
ParentProcessId: 1636
ParentImage: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
ParentCommandLine: "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -ep bypass
ParentUser: ATOMIC\Administrator

```

ProcessId: 5112
Image: C:\Windows\System32\HOSTNAME.EXE
FileVersion: 10.0.17763.1 (WinBuild.160101.0800)
Description: Hostname APP
Product: Microsoft® Windows® Operating System
Company: Microsoft Corporation
OriginalFileName: hostname.exe
CommandLine: "C:\Windows\system32\HOSTNAME.EXE"
CurrentDirectory: C:\Users\Administrator\
User: ATOMIC\Administrator
LogonGuid: {c5d2b969-17ff-63bc-42a1-070000000000}
LogonId: 0x7A142
TerminalSessionId: 2
IntegrityLevel: High
Hashes: MD5=7F95220A65A5A5D4A98873E86EF2E549,SHA256=1BFF2907C456F99277F45F9B2A21B1B3F11F6C01587D9E6D6F0B2B5F1472FE92,IMPHASH=
5CD891320C666621E9783444DB8CBA78
ParentProcessGuid: {c5d2b969-1967-63bc-9f00-000000001b01}
ParentProcessId: 1636
ParentImage: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
ParentCommandLine: "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" -ep bypass
ParentUser: ATOMIC\Administrator

```

Ignoring the first two entries, we are left with three logs generated by the emulated Atomic test. The following three entries are Process Create events (Sysmon Event ID 1) and a single Registry Value Set event (Sysmon Event ID 13).

Process Create Logs

```

UtcTime: 2023-01-09 15:05:50.984
ProcessGuid: {c5d2b969-2d4e-63bc-e800-000000001b01}
ProcessId: 200
Image: C:\Windows\System32\cmd.exe
FileVersion: 10.0.17763.1697 (WinBuild.160101.0800)
Description: Windows Command Processor
Product: Microsoft® Windows® Operating System
Company: Microsoft Corporation
OriginalFileName: Cmd.Exe
CommandLine: "cmd.exe" /c "REG ADD "HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Run" /V "Atomic Red Team" /t REG_SZ /F /D "C:\Path\AtomicRedTeam.exe""
CurrentDirectory: C:\Users\ADMINI~1\AppData\Local\Temp\2\
User: ATOMIC\Administrator
LogonGuid: {c5d2b969-17ff-63bc-42a1-070000000000}
LogonId: 0x7A142
TerminalSessionId: 2
IntegrityLevel: High
Hashes: MD5=911D039E71583A07320B32BDE22F8E22,SHA256=BC866CFCD0A37E24DC2634DC282C7A0E6F55209DA17A8FA105B07414C0E7C527,IMPHASH=
272245E2988E1E430500B852C4FB5E18
ParentProcessGuid: {c5d2b969-1967-63bc-9f00-000000001b01}
ParentProcessId: 1636

```

```

Image: C:\Windows\System32\reg.exe
FileVersion: 10.0.17763.1 (WinBuild.160101.0800)
Description: Registry Console Tool
Product: Microsoft® Windows® Operating System
Company: Microsoft Corporation
OriginalFileName: reg.exe
CommandLine: REG ADD "HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Run" /V "Atomic Red Team" /t REG_SZ /F /D "C:\Path\AtomicRedTeam.exe"
CurrentDirectory: C:\Users\ADMINI~1\AppData\Local\Temp\2\
User: ATOMIC\Administrator
LogonGuid: {c5d2b969-17ff-63bc-42a1-070000000000}
LogonId: 0x7A142
TerminalSessionId: 2
IntegrityLevel: High
Hashes: MD5=8A93ACAC33151793F8D52000071C0B06,SHA256=
19316D4266D0B776D9B2A05D5903D8CBC8F0EA1520E9C2A7E6D596086FA4DCAF,IMPHASH=BE482BE427FE212CFEF2CDA0E61F19AC
ParentProcessGuid: {c5d2b969-2d4e-63bc-e800-000000001b01}
ParentProcessId: 200
ParentImage: C:\Windows\System32\cmd.exe
ParentCommandLine: "cmd.exe" /c "REG ADD "HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Run" /V "Atomic Red Team" /t REG_SZ /F /D "C:\Path\AtomicRedTeam.exe""
ParentUser: ATOMIC\Administrator

```

Registry Value Set Log

```
Registry value set:  
RuleName: T1060.RunKey  
EventType: SetValue  
UtcTime: 2023-01-09 15:05:51.028  
ProcessGuid: {c5d2b969-2d4f-63bc-ea00-000000001b01}  
ProcessId: 552  
Image: C:\Windows\system32\reg.exe  
TargetObject: HKU\S-1-5-21-1966530601-3185510712-10604624-500\Software\Microsoft\Windows\CurrentVersion\Run\Atomic Red Team  
Details: C:\Path\AtomicRedTeam.exe  
User: ATOMIC\Administrator
```

Based on the images above, we can summarise the events generated by Atomic test T1547.001-1 with the following information:

- The test generated three significant events: two Process Create events and one Registry Value Set event.
- The two Process Create events are caused by the execution of the **reg.exe** binary via the **cmd.exe /c** parameter.
- For this type of attack, the **reg.exe** binary is being utilised by threat actors to modify the registry key **\SOFTWARE\Microsoft\Windows\CurrentVersion\Run**.
- Lastly, the Registry Value points to a malicious binary (**C:\Path\AtomicRedTeam.exe**), which indicates that the binary will be executed during user login.

Aurora EDR

Aurora is a Windows endpoint agent that uses Sigma rules and Indicators of Compromise (IOCs) to detect threat patterns on local event streams using Event Tracing for Windows (ETW). It is easy to correlate our previous learnings because Sigma rules are associated with the MITRE ATT&CK framework. To run Aurora, open a PowerShell window. Then navigate to **C:\Tools\Aurora** and execute **aurora-agent-64.exe**.

Administrator: C:\Windows\System32\cmd.exe

```
C:\Users\Administrator>cd C:\Tools\Aurora  
C:\Tools\Aurora> .\aurora-agent-64.exe
```

Wait for the message that states **Aurora Agent started** before going back to the execution of Atomic Test **T1547.001-2**. Once the tool has successfully loaded, proceed to the next Atomic Test.

Administrator: Windows PowerShell

```
PS C:\Users\Administrator> Invoke-AtomicTest T1547.001 -TestNumbers 2
```

```
PathToAtomicsFolder = C:\Tools\AtomicRedTeam\atomics
```

Executing test: T1547.001-2 Reg Key RunOnce

The operation completed successfully.

Done executing test: T1547.001-2 Reg Key RunOnce

After execution, you may observe that the window running Aurora EDR has generated several detections related to

Registry Modification

Note: You may ignore the detection of whoami.exe, as shown in the first two detection logs.

```
Jan 9 16:02:56 ATOMIC AURORA: Notice MODULE: Sigma MESSAGE: Sigma match found. RULE TITLE: CurrentVersion Autorun Keys Modification. RULE AUTHOR: Victor Sergeev, Danil Yugoslavskiy, Gleb Sukhodolskiy, Timur Zinniatullin, oscd.community, Tim Shelton, fracklas (copy). RULE DESCRIPTION: Detects modification of autostart extensibility point (ASEP) in registry. RULE FALSEPOSITIVES: Legitimate software automatically (mostly, during installation) sets up autorun keys for legitimate reason, legitimate administrator sets up autorun keys for legitimate reason. RULE ID: 20f0ee37-5942-4e45-b7d5-c5b5db9df5cd. RULE LEVEL: medium. RULE LINK: https://github.com/SigmaHQ/sigma/blob/0.22-1437-gd3dae24e9/rules/windows/registry/registry_set/registry_set_asep_reg_keys_modification_currentversion.yml. RULE MODIFIED: 2022/10/20. RULE PATH: public\windows\registry\registry_set\registry_set_asep_reg_keys_modification_currentversion.yml. RULE REFERENCES: https://github.com/redcanaryco/atomic-red-team/blob/f339e7da7d05f6057fdfcdd3742bfcf365fee2a9/atomics/T1547.001/T1547.001.md, https://docs.microsoft.com/en-us/sysinternals/downloads/autoruns, https://gist.github.com/GlebSukhodolskiy/0fc5fa5482903064b448890db1eaf9d, https://oddsvar.moe/2018/03/21/persistence-using-runonceex-hidden-from-autoruns-exe/. RULE SIGTYPE: public. COMPUTER: ATOMIC CORRELATION ACTIVITYID: {00000000-0000-0000-0000-000000000000} DETAILS: C:\Path\AtomicRedTeam\dl1 EVENTID: 13 EVENTTYPE: SetValue EXECUTION PROCESSID: 2388 EXECUTION_THREADID: 2188 IMAGE: C:\Windows\System32\reg.exe KEYWORDS: 0x8000000000000000 LEVEL: 4 MATCH_STRINGS: SetValue In EventID: \SOFTWARE\Microsoft\Windows\CurrentVersion\Run In TargetObject: \RunOnceEx\ PROVIDER_NAME: Microsoft-Windows-Sysmon RULENAME: {c5d2b969-3a8f-63bc-3e01-000000001b01} PROCESSID: 5420 PROVIDER_GUID: {5770385F-C22A-43E0-BF4C-06F5698FFB09} PROVIDER_NAME: Microsoft-Windows-Sysmon RULENAME: T1060_RunKey_SECURITY_USERID: S-1-5-18 TARGETOBJECT: HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\RunOnceEx\0001\Depend\1 TASK: 13 TIMECREATED_SYSTEMTIME: 2023-01-09T16:02:23.7563215Z USER: ATOMIC\Administrator UTIME: 2023-01-09 16:02:23.753 VERSION: 2 MINVERSION: 17763
```

```
Jan 9 16:02:54 ATOMIC AURORA: Notice MODULE: Sigma MESSAGE: Sigma match found. RULE TITLE: Reg Add Run Key. RULE AUTHOR: Florian Roth. RULE DESCRIPTION: Detects suspicious command line reg.exe tool adding key to RUN key in Registry. RULE FALSEPOSITIVES: Legitimate software automatically (mostly, during installation) sets up autorun keys for legitimate reasons., legitimate administrator sets up autorun keys for legitimate reasons., Discord. RULE ID: de587dce-915e-4218-aac4-835ca6af6f70. RULE LEVEL: medium. RULE LINK: https://github.com/SigmaHQ/sigma/blob/0.22-1437-gd3dae24e9/rules/windows/process_creation/proc_creation_win_reg_add_run_key.yml. RULE MODIFIED: 2022/10/09. RULE PATH: public\windows\process_creation\proc_creation_win_reg_add_run_key.yml. RULE REFERENCES: https://app.any.run/tasks/c6cf370c-867a-4314-b685-e101566766d7/, https://docs.microsoft.com/en-us/windows/win32/setupapi/run-and-runonce-registry-keys. RULE SIGTYPE: public. COMMANDLINE: REG ADD HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\RunOnceEx\0001\Depend /v 1 /d "C:\Path\AtomicRedTeam\dl1" COMPANY: Microsoft Corporation COMPUTER: ATOMIC CORRELATION ACTIVITYID: {00000000-0000-0000-0000-000000000000} CURRENTDIRECTORY: C:\Users\ADMINI~1\AppData\Local\Temp\2 DESCRIPTION: Registry Console Tool EVENTID: 1 EXECUTION PROCESSID: 2388 EXECUTION_THREADID: 2188 FILEVERSION: 10.0.17763.1 (WinBuild.160101.0800) HASHES: MD5-B8A93ACAC33151793F8052000071C0806_SHA256-1931604266D0877609B2A05D500308CB8F8EA1520E9C2A7E6D596086FAADCAF_IPHASH-BE482BE427FE212CFEF2CDA0E61F19AC IMAGE: C:\Windows\System32\reg.exe INTEGRITYLEVEL: High KEYWORDS: 0x8000000000000000 LEVEL: 4 LOGONGUID: {c5d2b969-17ff-63bc-42a1-070000000000} LOGONID: 0x7A142 MATCH_STRINGS: REG In CommandLine, 'ADD' In CommandLine, 'SOFTWARE\Microsoft\Windows\CurrentVersion\Run In CommandLine. OP CODE: 0 ORIGINALFILENAME: reg.exe PARENTCOMMANDLINE: "\"cmd.exe\" /c \"REG ADD HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\RunOnceEx\0001\Depend /v 1 /d \"C:\Path\AtomicRedTeam\dl1\"\"\" PARENTIMAGE: C:\Windows\System32\cmd.exe PARENTPROCESSGUID: {c5d2b969-3a8f-63bc-3e01-000000001b01} PARENTPROCESSID: 4172 PARENTUSER: ATOMIC\Administrator PROCESSID: {c5d2b969-3a8f-63bc-3e01-000000001b01} PRODUCT: \"Microsoft\Windows\Operating System\" PROVIDER_GUID: {5770385F-C22A-43E0-BF4C-06F5698FFB09} PROVIDER_NAME: Microsoft-Windows-Sysmon RULENAME: - SECURITY_USERID: S-1-5-18 TASK: 1 TERMINALSESSIONID: 2 TIMECREATED_SYSTEMTIME: 2023-01-09T16:02:23.6194978Z USER: ATOMIC\Administrator UTIME: 2023-01-09 16:02:23.611 VERSION: 5 MINVERSION: 17763
```

```
Jan 9 16:02:55 ATOMIC AURORA: Notice MODULE: Sigma MESSAGE: Sigma match found. RULE TITLE: Direct Autorun Keys Modification. RULE AUTHOR: Victor Sergeev, Danil Yugoslavskiy, oscd.community. RULE DESCRIPTION: Detects direct modification of autostart extensibility point (ASEP) in registry using reg.exe. RULE FALSEPOSITIVES: Legitimate software automatically (mostly, during installation) sets up autorun keys for legitimate reasons., legitimate administrator sets up autorun keys for legitimate reasons., Discord. RULE ID: 24357373-078F-44ed-9ac4-6d334a668a11. RULE LEVEL: medium. RULE LINK: https://github.com/SigmaHQ/sigma/blob/0.22-1437-gd3dae24e9/rules/windows/process_creation/proc_creation_win_susp_direct_asep_reg_keys_modification.yml. RULE MODIFIED: 2022/08/04. RULE PATH: public\windows\process_creation\proc_creation_win_susp_direct_asep_reg_keys_modification.yml. RULE REFERENCES: https://github.com/redcanaryco/atomic-red-team/blob/f339e7da7d05f6057fdfcdd3742bfcf365fee2a9/atomics/T1547.001/T1547.001.md. RULE SIGTYPE: public. COMMANDLINE: \"REG ADD HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\RunOnceEx\0001\Depend /v 1 /d \"C:\Path\AtomicRedTeam\dl1\"\"\" COMPANY: Microsoft Corporation COMPUTER: ATOMIC CORRELATION ACTIVITYID: {00000000-0000-0000-0000-000000000000} CURRENTDIRECTORY: C:\Users\ADMINI~1\AppData\Local\Temp\2 DESCRIPTION: Registry Console Tool EVENTID: 1 EXECUTION PROCESSID: 2388 EXECUTION_THREADID: 2188 FILEVERSION: 10.0.17763.1 (WinBuild.160101.0800) HASHES: MD5-B8A93ACAC33151793F8052000071C0806_SHA256-1931604266D0877609B2A05D500308CB8F8EA1520E9C2A7E6D596086FAADCAF_IPHASH-BE482BE427FE212CFEF2CDA0E61F19AC IMAGE: C:\Windows\System32\reg.exe INTEGRITYLEVEL: High KEYWORDS: 0x8000000000000000 LEVEL: 4 LOGONGUID: {c5d2b969-17ff-63bc-42a1-070000000000} LOGONID: 0x7A142 MATCH_STRINGS: ADD In CommandLine, \SOFTWARE\Microsoft\Windows\CurrentVersion\Run In CommandLine, \reg.exe In Image. OP CODE: 0 ORIGINALFILENAME: reg.exe PARENTCOMMANDLINE: "\"cmd.exe\" /c \"REG ADD HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\RunOnceEx\0001\Depend /v 1 /d \"C:\Path\AtomicRedTeam\dl1\"\"\" PARENTIMAGE: C:\Windows\System32\cmd.exe PARENTPROCESSGUID: {c5d2b969-3a8f-63bc-3e01-000000001b01} PARENTPROCESSID: 4172 PARENTUSER: ATOMIC\Administrator PROCESSID: {c5d2b969-3a8f-63bc-3e01-000000001b01} PRODUCT: \"Microsoft\Windows\Operating System\" PROVIDER_GUID: {5770385F-C22A-43E0-BF4C-06F5698FFB09} PROVIDER_NAME: Microsoft-Windows-Sysmon RULENAME: - SECURITY_USERID: S-1-5-18 TASK: 1 TERMINALSESSIONID: 2 TIMECREATED_SYSTEMTIME: 2023-01-09T16:02:23.6194978Z USER: ATOMIC\Administrator UTIME: 2023-01-09 16:02:23.611 VERSION: 5 MINVERSION: 17763
```

The results summarise the following information:

- Sigma Rule Title, Description, and Notable False Positives
- Sigma match strings used
- Execution details such as Process Names, Command-line parameters and Hashes

Based on the same events from Q1, what is the file name created by the test?

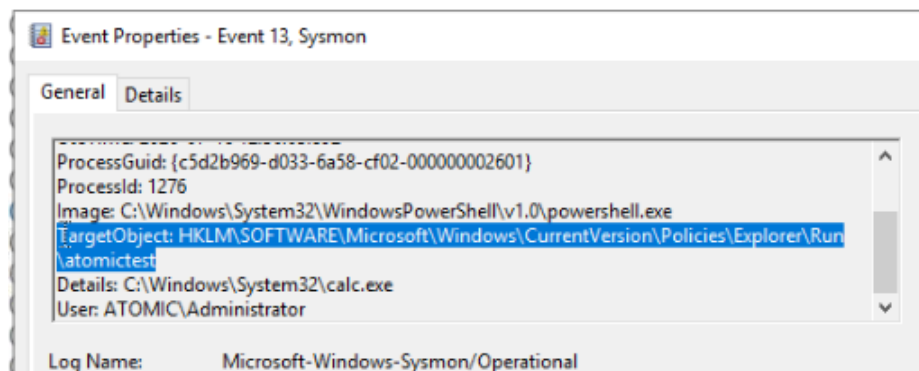
Operational Number of events: 18 (!) New events available				
Level	Date and Time	Source	Event ID	Task Category
Information	7/16/2026 12:30:18 PM	Sysmon	1	Process Crea...
Information	7/16/2026 12:30:18 PM	Sysmon	1	Process Crea...
Information	7/16/2026 12:30:18 PM	Sysmon	15	File stream c...
Information	7/16/2026 12:30:18 PM	Sysmon	11	File created (...)
Information	7/16/2026 12:30:18 PM	Sysmon	15	File stream c...
Information	7/16/2026 12:30:18 PM	Sysmon	11	File created (...)
Information	7/16/2026 12:30:18 PM	Sysmon	15	File stream c...
Information	7/16/2026 12:30:18 PM	Sysmon	11	File created (...)
Information	7/16/2026 12:30:18 PM	Sysmon	15	File stream c...
Information	7/16/2026 12:30:18 PM	Sysmon	11	File created (...)
Information	7/16/2026 12:30:17 PM	Sysmon	1	Process Crea...
Information	7/16/2026 12:30:16 PM	Sysmon	1	Process Crea...
Information	7/16/2026 12:30:16 PM	Sysmon	1	Process Crea...
Information	7/16/2026 12:29:55 PM	Sysmon	1	Process Crea...
Information	7/16/2026 12:29:19 PM	Sysmon	1	Process Crea...
Information	7/16/2026 12:28:43 PM	Sysmon	1	Process Crea...
Information	7/16/2026 12:28:06 PM	Sysmon	1	Process Crea...

Event 11, Sysmon		✕
General Details		
TargetFilename: C:\Users\Administrator\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup\vbsstartup.vbs CreationUtcTime: 2026-07-16 12:30:18.355		
Log Name:	Microsoft-Windows-Sysmon/Operational	
Source:	Sysmon	Logged: 7/16/2026 12:30:18 PM
Event ID:	11	Task Category: File created (rule: FileCreate)
Level:	Information	Keywords:
User:	SYSTEM	Computer: ATOMIC
OpCode:	Info	
More Information:	Event Log Online Help	

vbsstartup.vbs

Based on the Registry Value Set event generated after executing Atomic Test T1547.001-13, what is the value of the TargetObject field?

HKLM\Software\Microsoft\Windows\CurrentVersion\Policies\Explorer\Run\atomictest

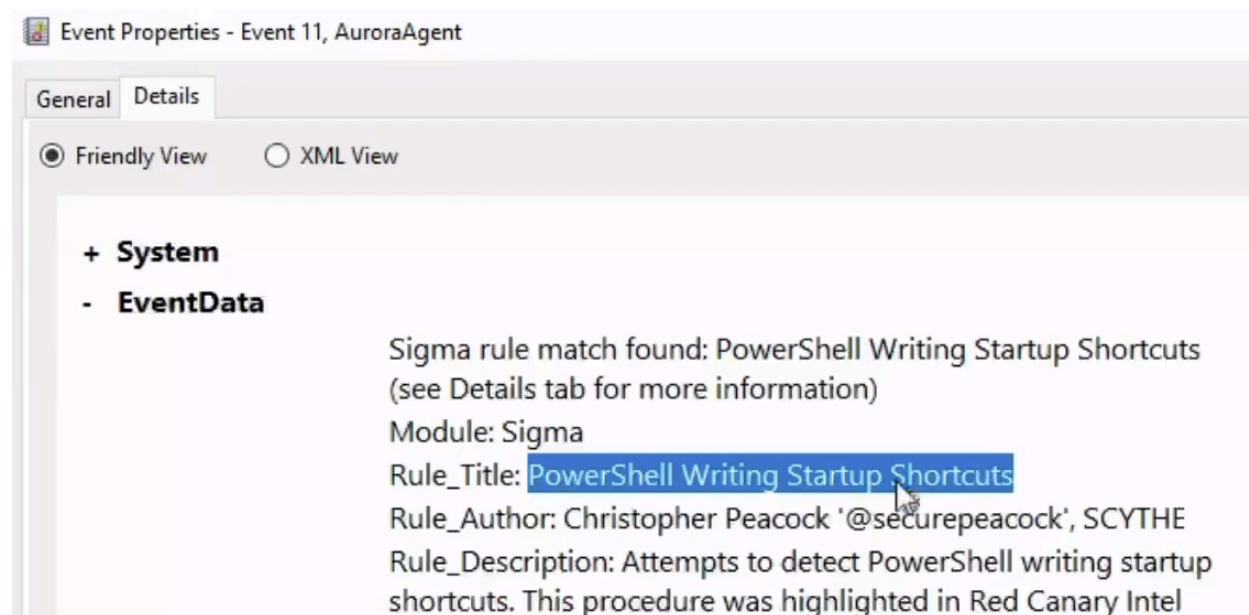
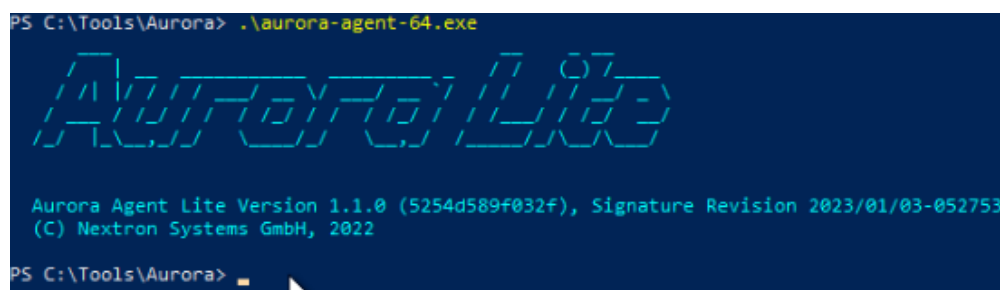


Excluding the WHOAMI detection, what is the title of the first rule triggered on Aurora EDR after executing Atomic Test T1547.001-7?

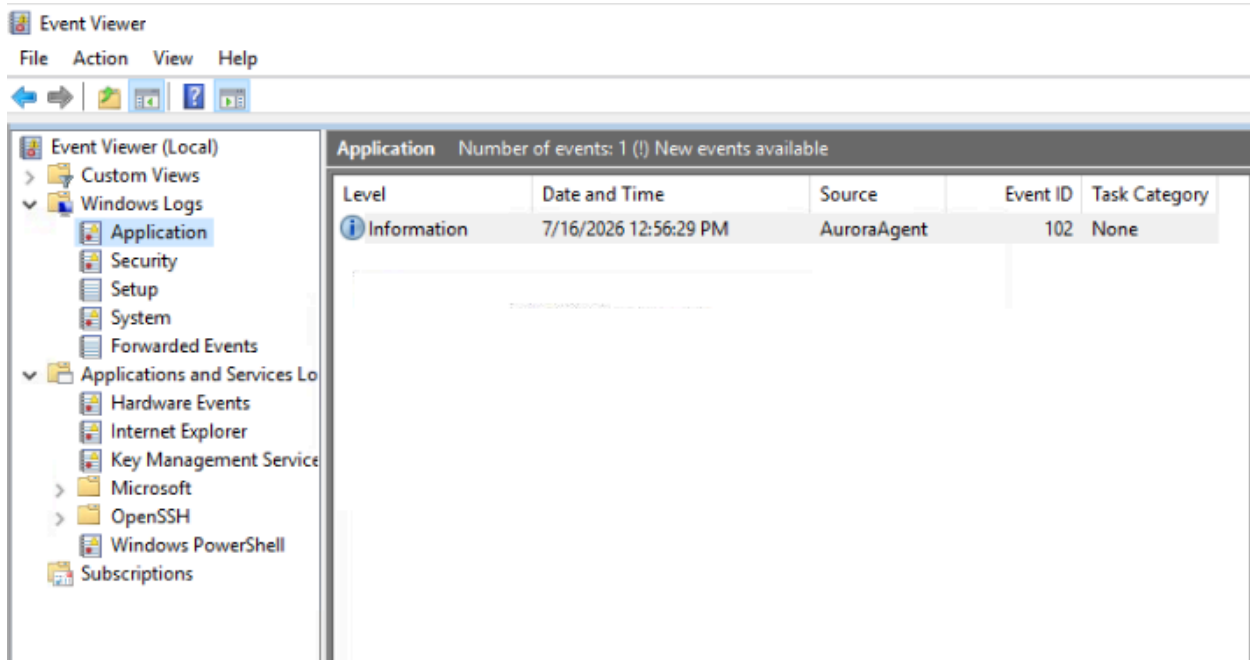
launch aurora

PS C:\Users\Administrator> cd C:\Tools\Aurora

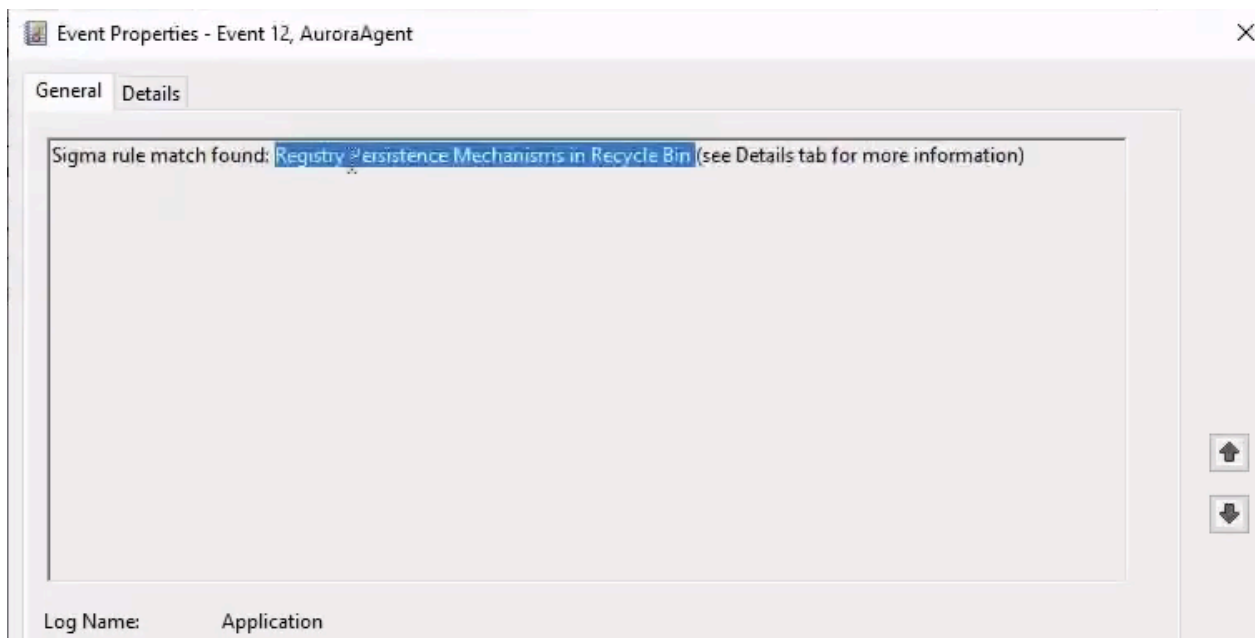
PS C:\Tools\Aurora> dir



PowerShell Writing Startup Shortcuts



Excluding the WHOAMI detection, what is the title of the first rule triggered on Aurora EDR after executing Atomic Test T1547.001-8?



Registry Persistence Mechanisms in Recycle Bin

Task 6 - Customising Atomic Red Team

In some cases, the Atomic Red Team cannot cater to the organisation's needs, such as emulating MITRE techniques that do not have a corresponding Atomic or tailoring a specific use case that only fits within a particular infrastructure or configuration setup. With that in mind, we need to be capable of creating custom Atomic tests and maximising tests via custom arguments.

Custom Input Arguments

T1136.001 - Create Account: Local Account

```
user@ATOMIC$ cat T1136.001/T1136.001.yaml
...
input_arguments:
  username:
    description: Username of the user to create
    type: String
    default: T1136.001_CMD
  password:
    description: Password of the user to create
    type: String
    default: T1136.001_CMD!
...
```

Using T1136.001 as an example, you may observe that the snippet above showcases two input names with the following information:

Input Name	Input Definition
username	• This input argument is described as the "Username of the user to create" • The username input type is String • The value defaults to T1136.001_CMD
password	• This input argument is described as the "Password of the user to create" • The username input type is String • The value defaults to T1136.001_CMD!

Executing this Atomic test will create a user named T1136.001_CMD, having T1136.001_CMD! as its password. We can test this out and verify it manually by executing Atomic Test #3 of T1136.001.

powershell.exe

```
PS C:\Users\Administrator> Invoke-AtomicTest T1136.001 -TestNumber  
s 3
```

```
PathToAtomicsFolder = C:\Tools\AtomicRedTeam\atomics
```

Executing test: T1136.001-3 Create a new user in a command prompt
The password does not meet the password policy requirements. Check
the minimum password length, password complexity and password history
requirements.

More help is available by typing NET HELPMSG 2245.

Done executing test: T1136.001-3 Create a new user in a command pr
ompt

powershell.exe

```
PS C:\Users\Administrator> net user
```

User accounts for \\ATOMIC

```
-----  
-----  
Administrator          DefaultAccount          Guest  
WDAGUtilityAccount
```

The command completed successfully.

Based on the output of the Atomic test and the `net user` command, the user was not created successfully due to the existing password policy requirement. We can try customising the input arguments and replacing the password to satisfy the policy. For this, we can either use one of these two parameters:

- **PromptForInputArgs** set the values of the input arguments interactively

powershell.exe

```
PS C:\Users\Administrator> Invoke-AtomicTest T1136.001 -TestNumber  
s 3 -PromptForInputArgs
```

- **InputArgs** - pass a hashtable that contains the key-value pair of input arguments and its values

powershell.exe

```
PS C:\Users\Administrator> $customArgs = @{ "username" = "THM_Atomic"; "password" = "p@ssw0rd" }
PS C:\Users\Administrator> Invoke-AtomicTest T1136.001 -TestNumbers 3 -InputArgs $customArgs
PathToAtomicsFolder = C:\Tools\AtomicRedTeam\atomics
```

Executing test: T1136.001-3 Create a new user in a command prompt
 The command completed successfully.
 Done executing test: T1136.001-3 Create a new user in a command prompt

The execution of the following commands below resulted in a new user named **THM_Atomic**.

The result can then be verified by checking the local users via **net.exe**. The user **THM_Atomic** was successfully created, as shown in the output below.

powershell.exe

```
PS C:\Users\Administrator> net user

User accounts for \\ATOMIC

-----
-----
Administrator          DefaultAccount          Guest
THM_Atomic              WDAGUtilityAccount
The command completed successfully.
```

Creating New Atomic Tests

The Invoke-AtomicRedTeam module also has a functionality that eases up the creation of Atomic Tests, and it is named the Atomic GUI. The Atomic GUI is a web-based form that users can fill out to generate the YAML definition of Atomic Tests. The result of this tool can then be inserted into the appropriate Atomic technique. To start the web application, execute **Start-AtomicGUI** via PowerShell.

powershell.exe

```
PS C:\Users\Administrator> Start-AtomicGui
```

Stopped all AtomicGUI Dashboards

Name	Port	Running	DashboardService
AtomicGUI	8487	True	UniversalDashboard.Services.DashboardService

Atomic Test Creation

Atomic Test Name

Atomic Test Description

Supported Platforms: ☒ Windows ☐ Linux ☐ macOS

Attack Commands

Executor for Attack Commands: PowerShell

☐ Requires Elevation to Execute Successfully?

Cleanup Commands (Optional)

GENERATE TEST DEFINITION YAML

ADD INPUT ARGUMENT (OPTIONAL)

ADD PREREQUISITE (OPTIONAL)

Executor for Prereq Commands: PowerShell

You may observe that all the values needed for a test can be quickly filled out, like:

- Atomic Test Name and Definition
- Supported Platforms
- Attack Commands, Command Executor, Required Elevation and Cleanup Commands
- Custom input arguments
- Prerequisites and their Executor

Atomic Test Creation

Atomic Test Name
Demo Atomic

Atomic Test Description
Demo

Supported Platforms: ☒ Windows ☐ Linux ☐ macOS

Attack Commands
echo A > C:\Test

Executor for Attack Commands
PowerShell

☐ Requires Elevation to Execute Successfully?

Cleanup Commands (Optional)
del C:\Test

GENERATE TEST DEFINITION YAML

ADD INPUT ARGUMENT (OPTIONAL)

ADD PREREQUISITE (OPTIONAL)

Executor for Prereq Commands
PowerShell

After writing the values, you can generate the output by clicking **GENERATE TEST DEFINITION**. This results in a YAML output that can be inserted into any Atomic Techniques needing modification.

Test Definition YAML

```
- name: Demo Atomic
  description: Demo
  supported_platforms:
  - windows
  executor:
    command: echo A > C:\Test
    cleanup_command: del C:\Test
    name: powershell
```

Note: Add two spaces to align the Atomic Tests in the target Atomic YAML file. You may click the highlighted button above twice to insert the appropriate indentation.

Answer the questions below

What parameter should you use to customise the input arguments interactively?

PromptForInputArgs - set the values of the input arguments interactively

What parameter should you use in conjunction with InputArgs/PromptForInputArgs to revert the changes made by the test?

cleanup

What is the default port used by the Atomic GUI?

AtomicGUI 8487 True UniversalDashboard

Task 7 - Case Study: Emulating APT37

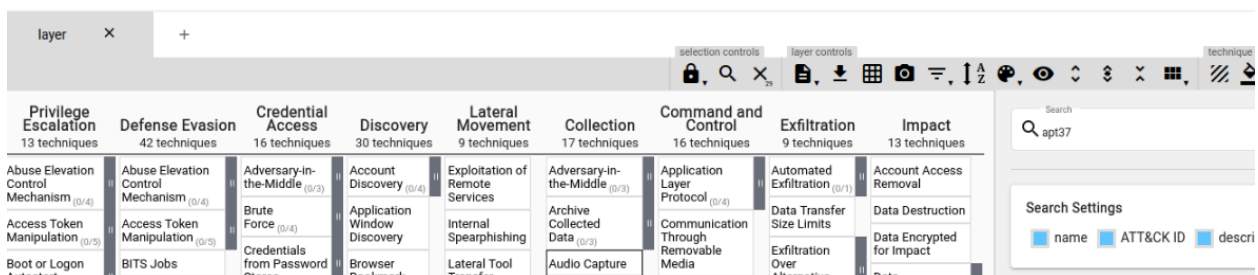
APT37, also known as Reaper, is a cyber espionage group that has been active since 2012 and is believed to be operating out of North Korea. The group has been known to target a wide range of organisations, including government agencies, defence contractors, and media companies.

You may follow these guidelines, which is a summary of the methodology covered above:

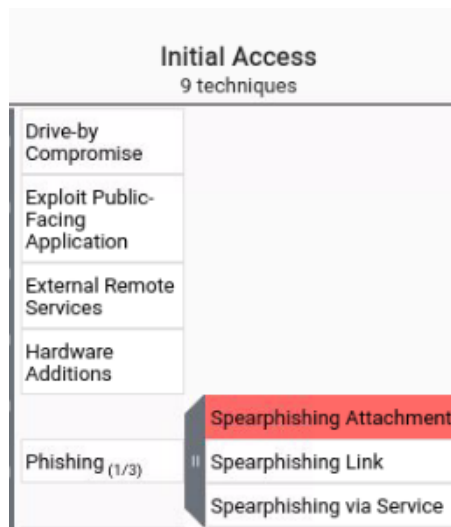
- Start by gathering the techniques attributed to APT37 using ATT&CK Navigator.
- Correlate all existing Atomics with the known techniques the given threat actor uses.
- Emulate all available Atomic Tests and observe the events generated via Event Viewer or Aurora EDR.
- Ensure to execute a cleanup after every test.
- Document and review the results.

Answer the questions below

Using the ATT&CK Navigator, how many techniques are attributed to APT37? 27



Using the mapping provided by the ATT&CK Navigator, what is the phishing technique used by the threat group?

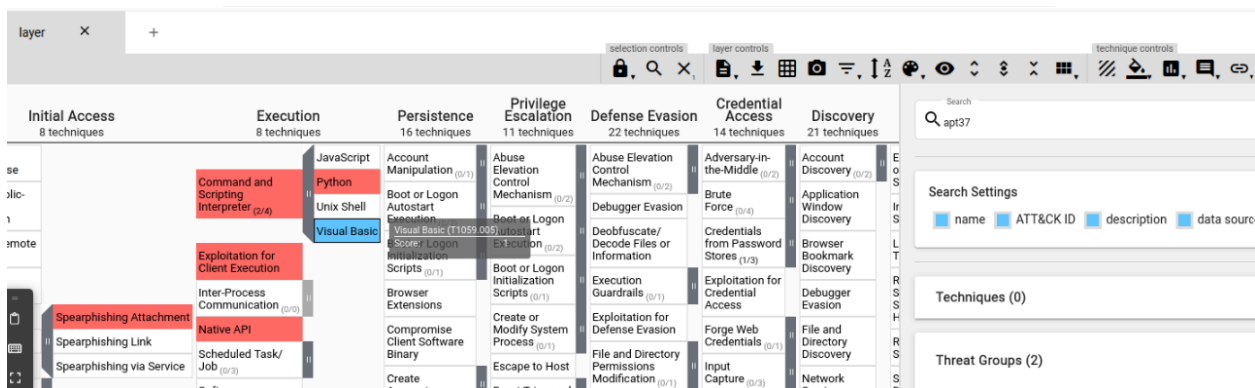


How many techniques attributed to APT37 have an existing Atomic file? 24

Is C:\Tools\AtomicRedTeam\atomics | Where-Object Name -Match "T1189|T1059|T1203|T1106|T1055|T1016|T1027|T1120|T1033|T1123|T1105|T1529"

Based on the results of Q3, which Atomic has no tests supported on Windows?
T1059.006

Choose only Linux (Layer controls in platform)



What is the description of the prerequisite needed for Atomic Test T1055-1?

Invoke-AtomicTest T1055-1 -ShowDetails

The 64-bit version of Microsoft Office must be installed

```

Dependencies:
Description: The 64-bit version of Microsoft Office must be installed
Check Prereq Command:
try {
    $wdApp = New-Object -COMObject "Word.Application"
    $path = $wdApp.Path
    Stop-Process -Name "winword"
    if ($path.contains("(x86)")) { exit 1 } else { exit 0 }
} catch { exit 1 }

```

How many Atomic tests have met the prerequisites for Atomic T1082? 15

Invoke-AtomicTest T1082 -CheckPrereqs

What are the three event IDs logged based on the execution of Atomic Test 1547.001-3? Provide the IDs in ascending order (e.g. 1,2,3).

Go to sysmon logs, clear

then run Invoke-AtomicTest 1547.001-3

```

PS C:\Users\Administrator> Invoke-AtomicTest T1547.001-3
PathToAtomicsFolder = C:\Tools\AtomicRedTeam\atomics

Executing test: T1547.001-3 PowerShell Registry RunOnce
Done executing test: T1547.001-3 PowerShell Registry RunOnce
PS C:\Users\Administrator>

```

Operational Number of events: 5				
Level	Date and Time	Source	Event ID	Task Category
Information	3/31/2025 4:05:45 AM	Sysmon	1	Process Cre...
Information	3/31/2025 4:05:45 AM	Sysmon	1	Process Cre...
Information	3/31/2025 4:05:45 AM	Sysmon	1	Process Cre...
Information	3/31/2025 4:05:46 AM	Sysmon	11	File created ...
Information	3/31/2025 4:05:46 AM	Sysmon	13	Registry val...

What command is executed (with default input value) by Atomic Test T1529-1? Do not run without the ShowDetails parameter.

Invoke-AtomicTest T1529-1

```

[*****BEGIN TEST*****]
Technique: System Shutdown/Reboot T1529
Atomic Test Name: Shutdown System - Windows
Atomic Test Number: 1
Atomic Test GUID: ad254fa8-45c0-403b-8c77-e00b3d3e7a64
Description: This test shuts down a Windows system.

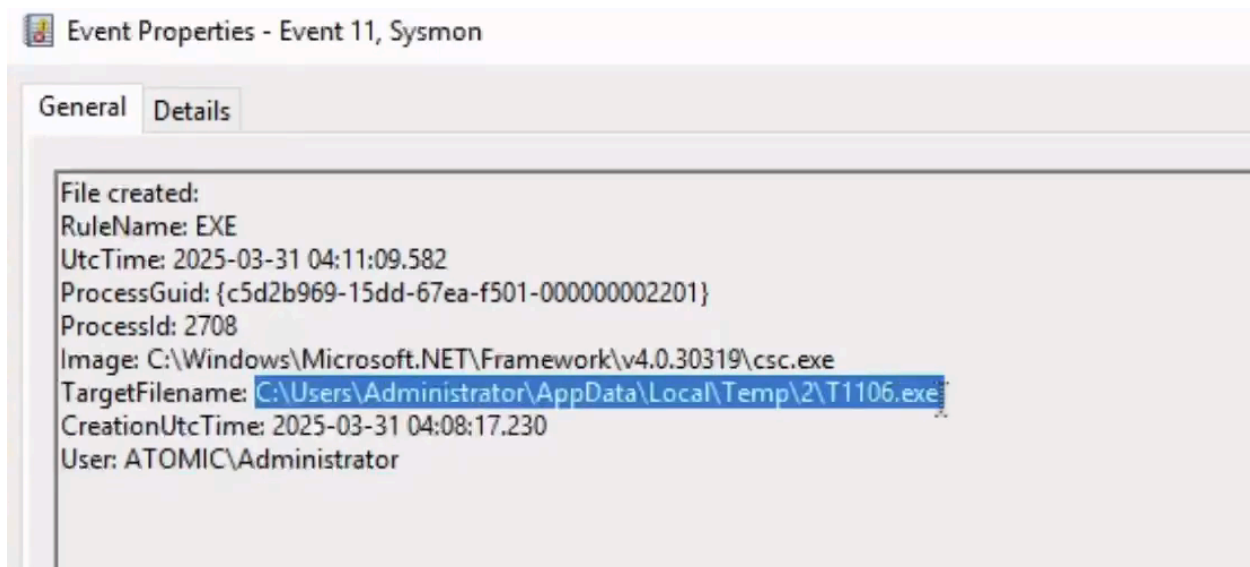
Attack Commands:
Executor: command_prompt
ElevationRequired: True
Command:
shutdown /s /t #{timeout}
Command (with inputs):
shutdown /s /t 1
[!!!!!!!END TEST!!!!!!!]

```

What is the value of the TargetFilename inside the File Creation log (Event ID 11) generated by Atomic Test T1106-1?

Invoke-AtomicTest T1106-1

C:\Users\Administrator\AppData\Local\Temp\2\T1106.exe



How many events are generated by executing the cleanup actions of Atomic T1105?

Invoke-AtomicTest T1105

Operational Number of events: 28				
Level	Date and Time	Source	Event ID	Task Category
 Information	3/31/2025 4:28:35 AM	Sysmon	1	Process Cre...
 Information	3/31/2025 4:28:34 AM	Sysmon	1	Process Cre...
 Information	3/31/2025 4:28:34 AM	Sysmon	1	Process Cre...
 Information	3/31/2025 4:28:35 AM	Sysmon	1	Process Cre...
 Information	3/31/2025 4:28:35 AM	Sysmon	1	Process Cre...
 Information	3/31/2025 4:28:35 AM	Sysmon	1	Process Cre...
 Information	3/31/2025 4:28:33 AM	Sysmon	1	Process Cre...
 Information	3/31/2025 4:28:32 AM	Sysmon	1	Process Cre...
 Information	3/31/2025 4:28:32 AM	Sysmon	1	Process Cre...
 Information	3/31/2025 4:28:34 AM	Sysmon	1	Process Cre...
 Information	3/31/2025 4:28:34 AM	Sysmon	1	Process Cre...

Task 8 - Conclusion

Throughout the room, we have tackled the following topics about how Blue Teamers can leverage Atomic Red Team:

- Breakdown of each Atomic, the main component of the Atomic Red Team Framework.
- Tools such as **Invoke-AtomicRedTeam** and **ATT&CK Navigator** to model and execute threat activity.
- Importance of reverting or cleaning up after conducting the tests.
- Logging and Detection Rules review based on the emulation activity.
- Customisation of Atomic tests for the specific needs during testing.
- Case study simulation to emulate the activity of a known threat group.

Threat Emulation Frameworks such as the Atomic Red Team are commonly viewed as a tool only for Red Teamers, yet being knowledgeable about it may aid in learning how threat actors do the job. Knowing how attackers do it simplifies understanding how to defend it.